

Standard Model of elementary particles

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Contents

1. Particle Classification
2. Fundamental Interactions and Bosons
3. Standard Model Formalism
4. Symmetries and Violations
5. Conclusion

Particle Classification

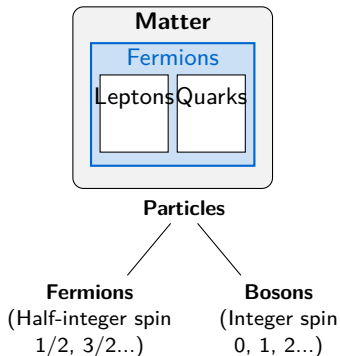
Classification Overview

Everything in the Standard Model is classified by Spin and Interactions [10, 9].

Type	Spin	Constituents	Force
Fermions	Half-integer	Matter	-
Bosons	Integer	Forces/Mass	-
Hadrons	-	Quarks (q)	Strong
Leptons	1/2	Fundamental	Weak/EM

Hadrons Sub-groups

- **Baryons** (qqq): Fermions (e.g., Proton, Neutron).
- **Mesons** ($q\bar{q}$): Bosons (e.g., Pion, Kaon).



Fundamental Fermions: Leptons

Leptons are fundamental particles that do not interact via the Strong force. Neutrinos are neutral and interact ONLY via the Weak force.

Particle	Mass	Charge	Spin	Lifetime / Stability	Main Decay	Discovery
Electron (e^-)	0.511 MeV	-1	1/2	Stable ($> 10^{26}$ yrs)	-	1897 (J.J. Thomson)
Muon (μ^-)	106 MeV	-1	1/2	$2.2\mu s$	$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$	1937 (Anderson/Neddermeyer)[1]
Tau (τ^-)	1.77 GeV	-1	1/2	10^{-13} s	$\tau^- \rightarrow \text{hadrons} + \nu_\tau$	1975 (M. Perl)
e Neutrino (ν_e)	< 1 eV	0	1/2	Stable	-	1956 (Cowan/Reines)
μ Neutrino (ν_μ)	< 1 eV	0	1/2	Stable	-	1962 (Lederman/Schwartz)
τ Neutrino (ν_τ)	< 1 eV	0	1/2	Stable	-	2000 (DONUT Collab.)

Antimatter: Every lepton has an antiparticle (e.g., the Positron e^+ discovered in 1932 [1]).

Neutrino Chirality: Neutrinos are extremely light and, in the Standard Model, exist ONLY as **Left-Handed** chiral states. Trillions pass through us every second unhindered.

Fundamental Fermions: Quarks

Quarks are fractionally charged fermions (proposed by Gell-Mann in 1964 [4]) that carry **Colour** charge, meaning they interact via the Strong force. They are never found isolated (Confinement).

Flavour	Mass	Charge	Spin	Components / Notes	Discovery
Up (u)	2 – 3 MeV	+2/3	1/2	Protons (uud) and Neutrons (udd)	1968 (SLAC)
Down (d)	4 – 6 MeV	-1/3	1/2	Protons (uud) and Neutrons (udd)	1968 (SLAC)
Charm (c)	1.4 GeV	+2/3	1/2	J/ψ resonance ($c\bar{c}$)	1974 (Predicted by GIM [5])
Strange (s)	100 MeV	-1/3	1/2	Found in Kaons. Slow Weak decay	1968 (SLAC)
Top (t)	~ 172 GeV	+2/3	1/2	Heaviest! Decays before hadronizing	1995 (Fermilab)
Bottom (b)	4.5 GeV	-1/3	1/2	Key for CP violation studies	1977 (Fermilab)

Quark Properties: Flavours and Colours

Flavours

"Flavour" simply identifies the species of the quark (u, d, c, s, t, b). The Weak force is the only interaction capable of changing a quark's flavour (e.g., $d \rightarrow u$ in β -decay).

Colours ($SU(3)_C$)

Quarks carry a "Colour" charge (Red, Green, Blue). This was introduced to satisfy the Pauli Exclusion Principle in baryons like $\Delta^{++}(uuu^\uparrow)$, ensuring the overall wavefunction is antisymmetric.

Colour Confinement

Only "Colourless" states (Singlets) are observable in nature. If you try to pull two quarks apart, the strong force energy creates a new $q\bar{q}$ pair instead of isolating them (Hadronization/Jets).

- **Baryons (qqq):** *Red + Green + Blue = White.*
- **Mesons ($q\bar{q}$):** *Colour + Anticolour = White.*

Fundamental Interactions and Bosons

Fundamental Interactions and Bosons

Interactions are mediated by the exchange of **Gauge Bosons** (Spin 1). The Higgs is a special Spin 0 boson providing mass.

Force	Mediator Boson	Mass	Charge	Spin	Range	Couples to...
Strong	Gluon (g) (8 types)	0	0	1	10^{-15}m	Color Charge (Quarks)
Electromagnetic	Photon (γ)	0	0	1	∞	Electric Charge
Weak	$W^{\pm}, Z^0$	80.4, 91.2 GeV	$\pm 1(W), 0(Z)$	1	10^{-18}m	Flavor/Isospin
Gravity	<i>Graviton?</i>	0	0	2	∞	Energy/Mass
Mass Mechanism	Higgs (H^0)	125 GeV	0	0	Local	Fermions & W, Z

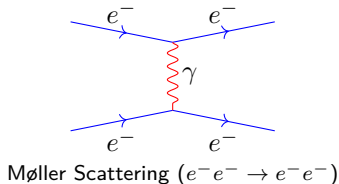
Yukawa Principle

The range of a force is inversely proportional to the mass of its mediator: $R \sim \frac{\hbar}{Mc}$. Weak force is short-range because W, Z are heavy!

Electromagnetic and Strong Interactions

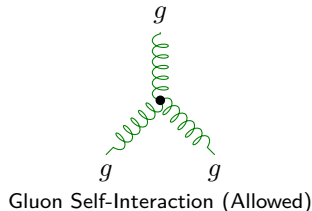
Quantum Electrodynamics (QED)

- **Mediator:** Photon (γ). Massless, Abelian.
- **Coupling (α):** *Increases* at high energy due to vacuum screening of virtual pairs.
- Conserves Flavor, Charge, Color.



Quantum Chromodynamics (QCD)

- **Mediators:** 8 Gluons (g). Non-Abelian.
- **Coupling (α_s):** *Decreases* at high energy (**Asymptotic Freedom**).
- **Self-Interaction:** Gluons carry color and attract each other (Confinement).

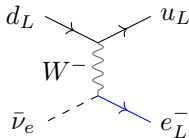


Weak Interaction

The Weak force is the only interaction that changes Particle Flavour and violates Parity. At high energies, it unifies with EM into the Electroweak force [12].

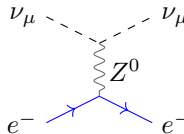
Charged Current ($W^\pm$)

- **Mass:** ≈ 80.4 GeV (Short range).
- Couples ONLY to **Left-Handed** fermions. Right-handed particles don't feel it!
- **Example:** β -Decay ($d \rightarrow uW^-$).



Neutral Current (Z^0)

- **Mass:** ≈ 91.2 GeV (Short range).
- No flavor change ($q \rightarrow qZ$).
- Couples to both L and R chiralities.
- **Example:** $\nu_\mu e^- \rightarrow \nu_\mu e^-$.



Higgs Boson (H^0)

The Mass Problem

Gauge symmetries strictly forbid bare mass terms for W, Z bosons and chiral fermions in the Lagrangian. Yet, they have mass!

Solution: Spontaneous Symmetry Breaking (SSB) via the Higgs Field.

- We introduce a complex scalar field ϕ with a "Mexican Hat" potential:
$$V(\phi) = \mu^2|\phi|^2 + \lambda|\phi|^4.$$
- The minimum of the potential is not at zero, so the field acquires a **Vacuum Expectation Value (VEV)**: $v \approx 246$ GeV.
- **Mechanism**: As particles interact with this background field v , they acquire inertia/mass.
- **Bosons**: W and Z become massive. The photon remains massless.
- **Fermions**: Acquire mass via Yukawa couplings ($m_f \propto y_f v$).

The scalar excitation of this field is the **Higgs Boson** (LHC in 2012, $m \approx 125$ GeV).[6]

Standard Model Formalism

Gauge Groups and Covariant Derivative

The Standard Model is a Quantum Field Theory based on the local gauge symmetry group [11, 12]:

$$G_{SM} = SU(3)_C \times SU(2)_L \times U(1)_Y$$

To maintain Local Gauge Invariance $\psi \rightarrow e^{i\alpha(x)}\psi$, the standard derivative ∂_μ is replaced by the **Covariant Derivative** D_μ , which contains all the force interactions:

$$D_\mu = \partial_\mu - ig' \frac{Y}{2} B_\mu - ig \frac{\tau^i}{2} W_\mu^i - ig_s \frac{\lambda^a}{2} G_\mu^a$$

Group	Coupling	Fields / Generators	Acts on...
$U(1)_Y$	g'	B_μ (1) / Y	Hypercharge
$SU(2)_L$	g	$W_\mu^{1,2,3}$ (3) / τ^i (Pauli)	Weak Isospin (Left-Handed Fermions)
$SU(3)_C$	g_s	$G_\mu^{1\dots 8}$ (8) / λ^a (Gell-Mann)	Color Charge (Quarks)

Standard Model Lagrangian ($\mathcal{L}_{SM}$)

The master equation describing all fundamental interactions (except gravity).

$$\mathcal{L}_{SM} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{I. Gauge}} + \underbrace{i\bar{\psi}\not{D}\psi}_{\text{II. Matter}} + \underbrace{|D_{\mu}\phi|^2 - V(\phi)}_{\text{III. Higgs}} - \underbrace{y_{ij}\bar{\psi}_L\phi\psi_R}_{\text{IV. Yukawa}}$$

I. Gauge Sector (Kinetic): Describes force carriers. For Non-Abelian forces (W, G), $F_{\mu\nu}$ contains a commutator $+g[A_{\mu}, A_{\nu}]$ causing self-interactions.

II. Matter Sector (Dirac): Kinematics of Quarks and Leptons. $\not{D} = \gamma^{\mu}D_{\mu}$. Fermions are intrinsically massless here!

III. Higgs Sector (EWSB): Kinetic term $|D_{\mu}\phi|^2$ generates mass for W, Z when $\phi \rightarrow v$. $V(\phi)$ drives symmetry breaking.

IV. Yukawa Sector: Links Left (ψ_L) and Right (ψ_R) fields via the Higgs, generating fermion masses $m_f = y_f v / \sqrt{2}$ and mixing matrices.

Symmetries and Violations

Discrete Symmetries: C, P, T

- **Parity (P) - Spatial Inversion:** Acts like a mirror. It flips spatial coordinates and momentum ($\vec{p} \rightarrow -\vec{p}$), but leaves spin unchanged. $(x, y, z) \rightarrow (-x, -y, -z)$
- **Charge Conjugation (C):** Swaps particles with their antiparticles. It flips all internal quantum numbers (electric charge, color) but keeps mass and spin intact. $e^- \rightarrow e^+$, $q \rightarrow \bar{q}$
- **Time Reversal (T):** Inverts the flow of time, effectively running processes backward. It reverses both momentum and spin. $t \rightarrow -t$

CPT Theorem

Any Lorentz-invariant local QFT must be exactly invariant under the combined operation of C, P, and T.

Implications:

- Particles and Antiparticles must have the **exact same mass and lifetime**.
- A universe made of antimatter, mirrored, and running backward in time would obey the exact same physical laws!

Helicity and Chirality (Left vs Right)

The chirality defines particle "handedness" [10].

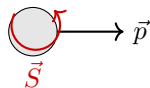
- **Helicity (h):** The projection of a particle's Spin ($\vec{S}$) along its momentum ($\vec{p}$). $h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$
- **Right-Handed:** Spin points in the **SAME** direction as motion.
- **Left-Handed:** Spin points **OPPOSITE** to motion.

Parity Connection

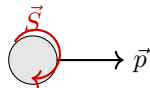
Under Parity (P), spatial directions reverse ($\vec{p} \rightarrow -\vec{p}$), but Spin (an axial vector) does not ($\vec{S} \rightarrow \vec{S}$).

Consequence: A mirror reflection flips a Left-Handed particle into a Right-Handed one!

Left Handed



Right Handed



Parity Violation (V-A Theory)

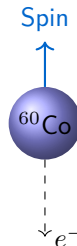
The Weak force is unique: it discriminates based on "Handedness" (Chirality).

V-A Interaction

The W boson couples **only to Left-Handed** fermions. Since the mirror reflection (Parity) of a Left-state is a Right-state (which doesn't interact), **Parity is maximally violated**.

Wu Experiment (1957): [13]

- ^{60}Co nuclei were aligned by a B-field (Spin $\uparrow$).
- **Observation:** In β -decay, electrons were emitted preferentially opposite to the spin direction.
- **Conclusion:** If Parity were conserved, emission would be symmetric (Up = Down). The asymmetry proved P-violation.



Quark Mixing: CKM Matrix

Mass eigenstates (d, s, b) are not the same as Weak eigenstates (d', s', b'). Quarks "mix" across generations when interacting with the W boson.

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \underbrace{\begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}}_{\text{CKM Matrix}} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

Hierarchy:

Quarks prefer to stay in their generation ($V_{tb} \approx 1$). Off-diagonal terms (Cabibbo Angle) allow rare flavor changes.

CP Violation

To have CP Violation, the mixing matrix must have an imaginary phase δ . [7]

- **3 Generations** are required mathematically to allow this complex phase!
- Discovered in Neutral Kaon decays ($K_L \rightarrow \pi\pi$) in 1964.[2]
- Consequence: the universe is predominantly made of matter.

Lepton Mixing: Neutrino Oscillations and PMNS Matrix

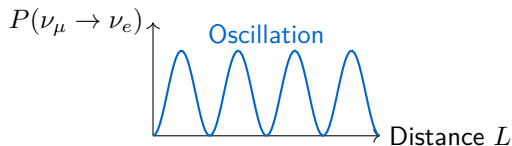
Neutrinos change flavor ($\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$) as they travel [8, 3].

- Neutrinos are *produced* and *interact* as **Flavor states** (ν_e, ν_μ, ν_τ) via the Weak force.
- However, they *propagate* through space as **Mass states** (ν_1, ν_2, ν_3).
- **Implication:** Since mass states have different masses ($\Delta m^2 \neq 0$), they propagate at slightly different velocities. This creates a quantum phase shift, causing the flavor to oscillate! $\Rightarrow$ **Neutrinos have mass!**

PMNS Matrix: Connects the two sets of states.

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{PMNS} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

Difference from CKM: Mixing angles in PMNS are extremely **LARGE**. For example, the mass state ν_2 is a near-equal mix of all three flavors.



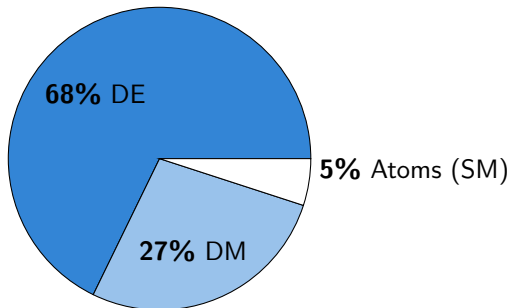
Oscillation probability depends on L/E and Δm^2 .

Conclusion

Unknown Universe

"The Standard Model is the most successful theory in history, yet we know it must be wrong."
It describes ordinary matter perfectly, but that's only a tiny fraction of the cosmos.

- **Dark Matter (27%):** Invisible mass holding galaxies together (evident from rotational curves).
- **Dark Energy (68%):** Unknown energy driving the accelerated expansion of the universe.
- **SM Matter (5%):** Everything we have studied in particle physics!



Open questions:

- Hierarchy Problem ($M_H \ll M_{Pl}$).
- Why is there Matter/Antimatter Asymmetry?
- Gravity is not included in the SM.

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